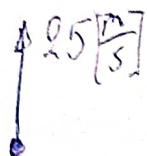


F1



A small blue dot representing an object is shown with a vertical arrow pointing upwards from it. Next to the arrow is the text $25 \frac{m}{s}$.

$$y = y_0 + v_{0y} t - \frac{1}{2} g t^2$$

$$0 = 25 - \frac{1}{2} g t^2$$

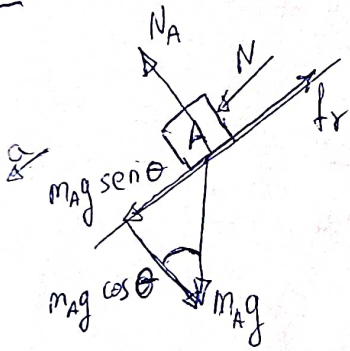
$$t = \frac{50}{g} = 5 [s]$$

$$x = x_0 + v_{0x} t + \frac{1}{2} a t^2$$

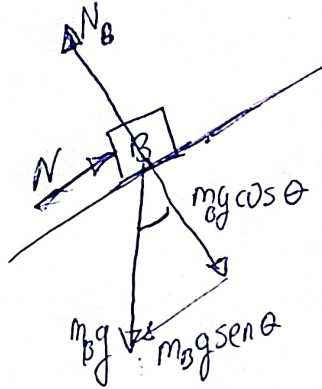
$$x = \frac{1}{2} (2) (5)^2$$

$$x = 25 [m]$$

F2



$$\begin{cases} N_A - m_A g \cos \theta = 0 \\ m_A g \sin \theta - f_r + N = m_A a \end{cases}$$



$$\begin{cases} N_B - m_B g \cos \theta = 0 \\ m_B g \sin \theta - N = m_B a \end{cases}$$

$$m_B g \sin \theta - N = m_B \left(\frac{m_A g \sin \theta - f_r + N}{m_A} \right)$$

$$\cancel{m_B g \sin \theta} - N = \cancel{m_B g \sin \theta} = \frac{m_B}{m_A} f_r + \frac{m_B}{m_A} N$$

$$\frac{m_B}{m_A} \mu_r N_A = N \left(\frac{m_B}{m_A} + 1 \right)$$

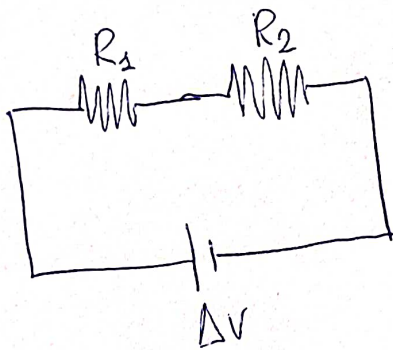
$$N = \frac{\frac{m_B}{m_A} \mu_r \cancel{m_A g \cos \theta}}{\frac{m_B}{m_A} + 1} = \frac{\mu_r m_B g \cos \theta}{1 + \frac{m_B}{m_A}} = 0.8 [N]$$

F3

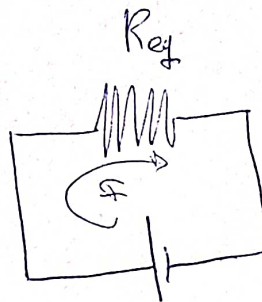
$$V_i = \frac{k_e q_1}{a} + \frac{k_e q_2}{2a} = 0$$

$$q_2 = -2q_1 \rightarrow q_1 = -2 [4 \mu\text{C}] = \underline{\underline{-8 [\mu\text{C}]}}$$

F4



=



$$R = \rho \frac{L}{A}$$

$$R_{eq} = R_1 + R_2$$

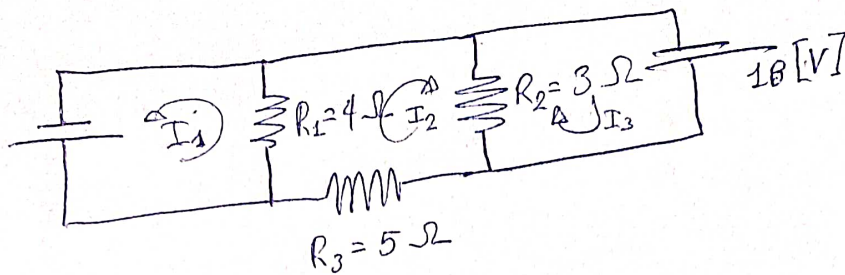
$$\Rightarrow R_{eq} = \frac{4L}{\pi D^2} (\rho_1 + \rho_2)$$

$$\Delta V = R_{eq} I$$

$$I = \frac{\Delta V}{R_{eq}}$$

$$I = \underline{4,02 \text{ [A]}}$$

F5



$$\sum V = 0$$

$$8 - 4(I_1 + I_2) = 0 \rightarrow I_1 = \frac{8 - 4I_2}{4} = 2 - I_2$$

$$-4(I_2 + I_1) - 3(I_2 - I_3) - 5I_2 = 0$$

$$18 - 3(I_3 - I_2) = 0 \rightarrow 18 = -3(I_2 - I_3)$$

$$-4(I_2 + I_1) + 18 - 5I_2 = 0$$

$$-9I_2 - 4I_1 + 18 = 0$$

$$-9I_2 - 4(2 - I_2) + 18 = 0$$

$$-5I_2 - 8 + 18 = 0$$

$$I_2 = -\frac{10}{5} = -2 \text{ [A]}$$